Wearables at Scale:

Technical progress and practical barriers for measuring physical activity in national population health surveillance

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Geneva, Switzerland

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Health Research from Home Connect, London, U.K.

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The Problem

2

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WHO: Core functions and responsibility

- Provide leadership on global health matters
- **Monitor and assess health trends (diseases, determinants, policy)**
- **Set global norms and standards (including guidelines & guidance)**
- Articulate evidence-based policy options
- Provide technical support to countries
- **Shape the health research agenda**

4

What is Public Health?

- Organised effort to protect and improve the nation's health, prevent disease, and reduce health inequalities, *focusing on populations* rather than individuals, through prevention, health promotion, and health protection measures

Faculty of Public Health UK

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What is population surveillance?

Public health surveillance is the ongoing, systematic collection, analysis and interpretation of data with prompt dissemination of results to those who need to know, particularly those who are in a position to take action.

- Monitors health status, disease and associated risk factors in the whole population
- Provides estimates and track trends over time
- Inform health care planning and policy development
- Enables evidence-based decision making (and resource allocation!)
- It measures both the need for interventions and the effect of interventions
- It is a core public heath function

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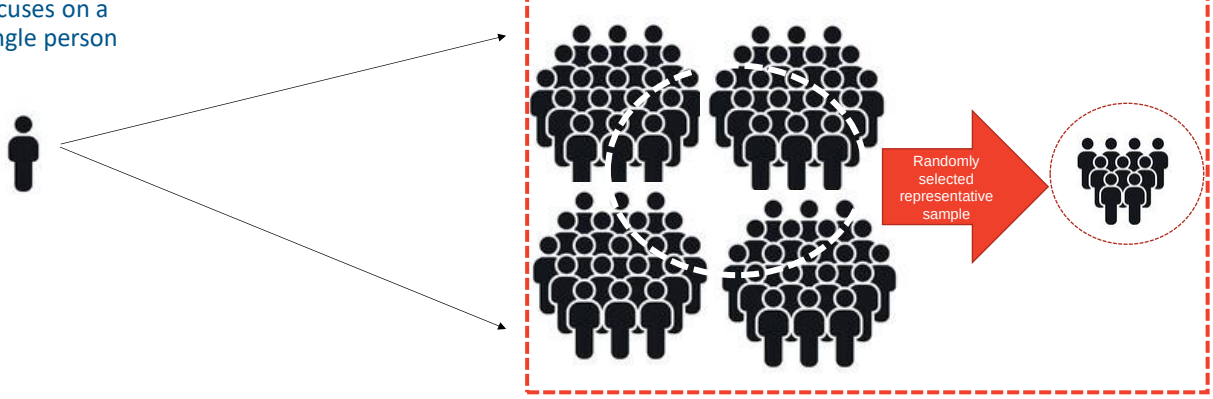
Measuring physical activity at different scales

Individual-level measurement focuses on a single person

versus

Population-level measurement uses aggregate data from a large group to understand trends and overall characteristics.



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Global monitoring - global comparisons

194 countries

8

What are we trying to measure in population health surveillance?

9

Global Action Plan On Physical Activity 2018-2030
(Adopted in World Health Assembly Resolution WHA71.6)

There are many ways to be active – walking, cycling, sport, active recreation, dance and play - and many policy opportunities to increase participation.

GLOBAL TARGETS

BY 2025
10%

BY 2030
15%

And to be more precise…..

A 15% relative reduction in the global prevalence of ***insufficient physical activity*** in adults and adolescents by 2030

Where….

insufficient physical activity = not meeting global recommendation

To access visit www.who.int/health-topics/physical-activity

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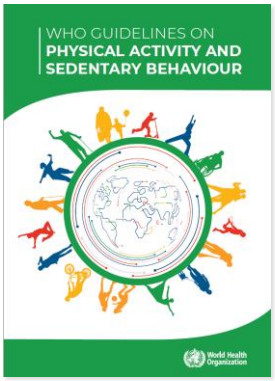
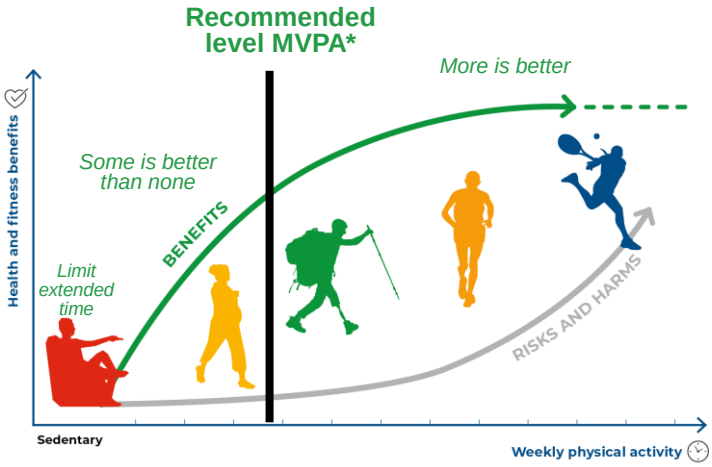
Global recommendations on physical activity (adults)





* Moderate- to vigorous- intensity physical activity

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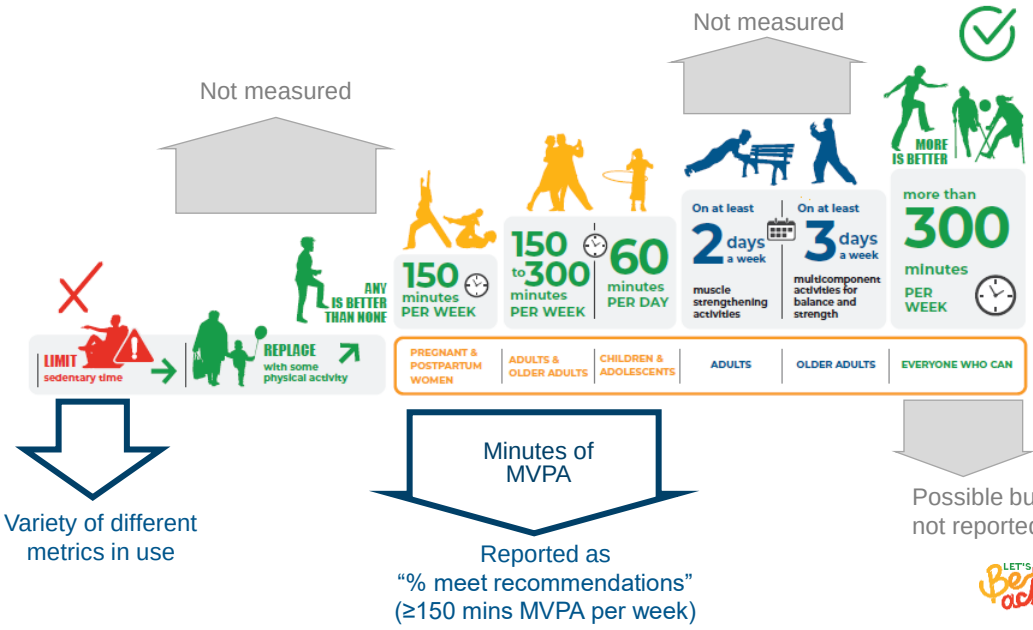
What we need to measure ?



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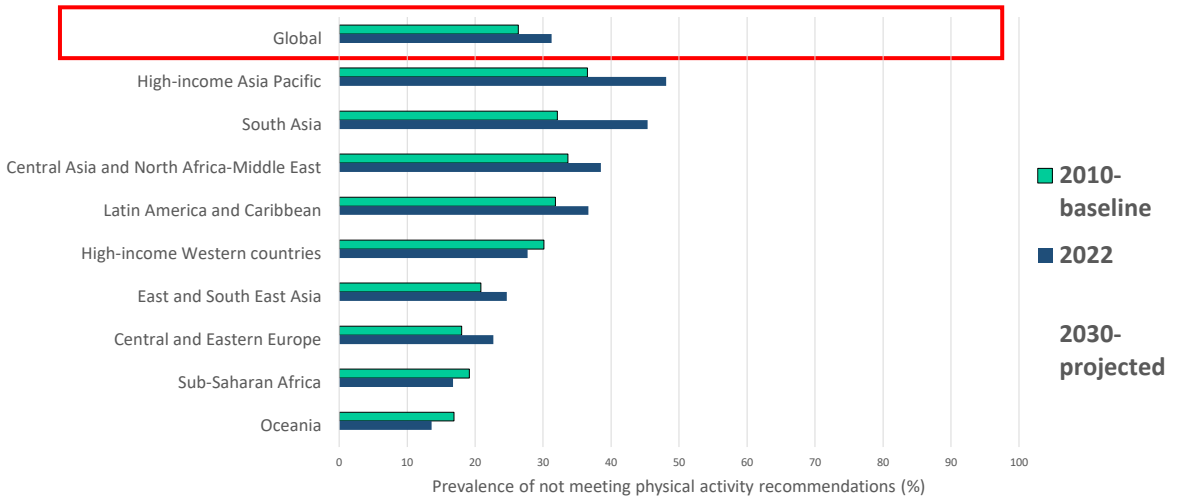
Prevalence of physical inactivity globally and by country groupings for years 2010, 2022 and 2030



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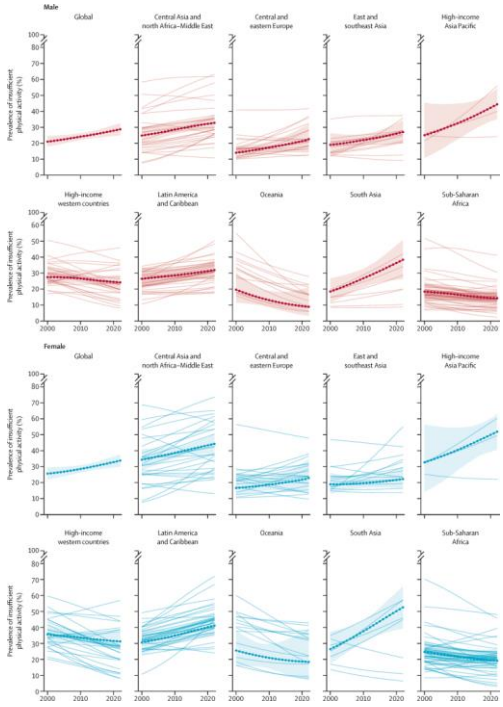

National, regional, and global trends in insufficient physical activity among adults from 2000 to 2022: a pooled analysis of 507 population-based surveys with 5·7 million participants

*Tessa Strain, Seth Flaxman, Regina Guthold, Elizaveta Semenova, Melanie Cowan, Leanne M Riley, Fiona C Bull, Gretchen A Stevens, and the Country Data Author Group**

Strain et al., Lancet Global Health 2024

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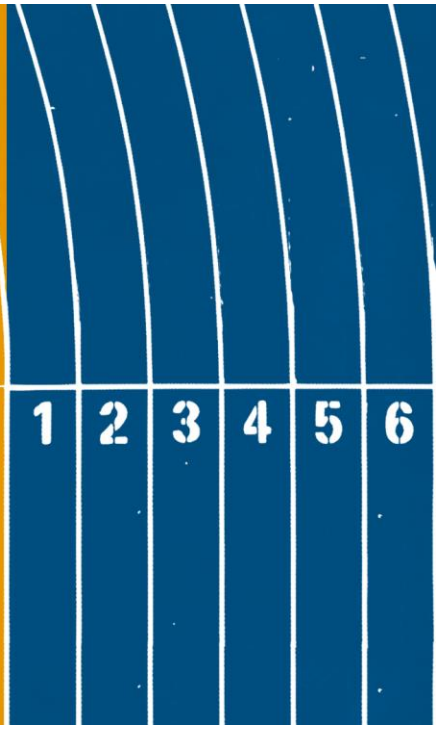
The world is off track!

Most countries will not meet the global target to reduce physical inactivity by 15% by 2030.



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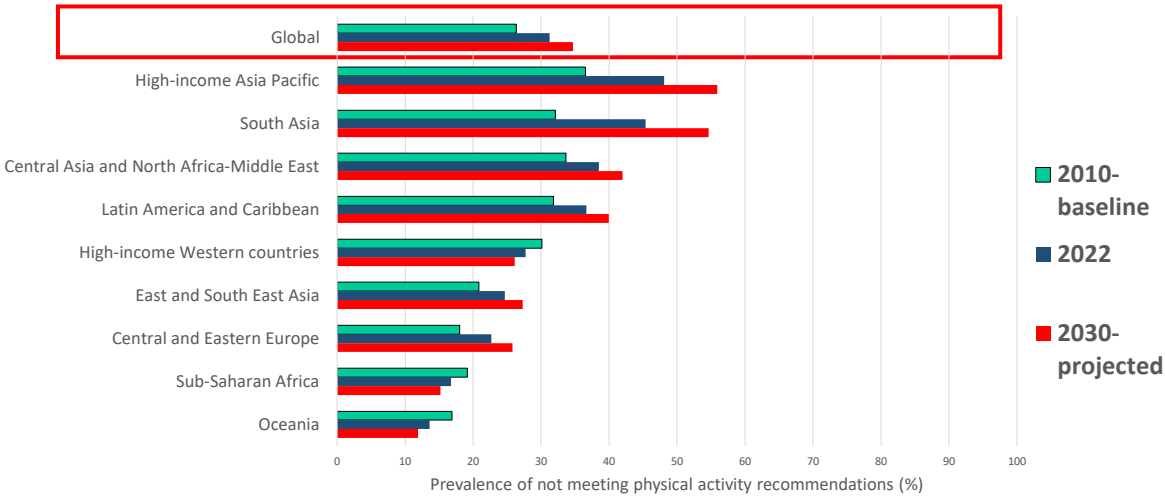
Prevalence of physical inactivity globally and by country groupings for years 2010, 2022 and 2030



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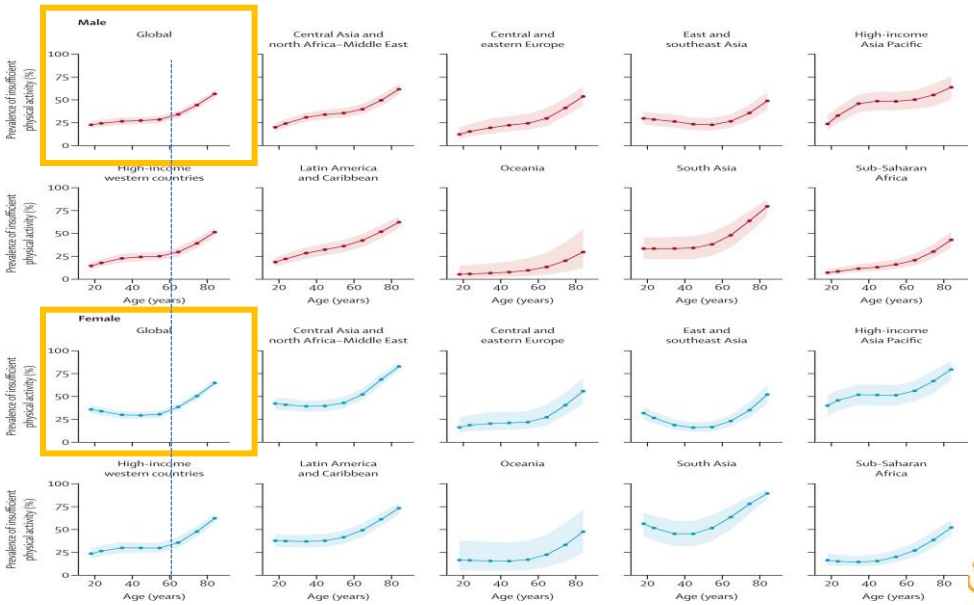
Prevalence of physical inactivity globally and by country groupings, by sex and by age; 2022



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After 60 years of age physical inactivity rises rapidly

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What instruments are we using now?

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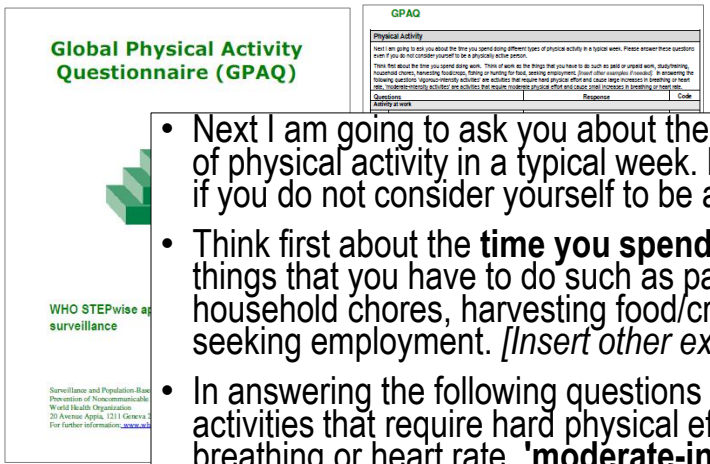
Self report questionnaire: Example GPAQ



- Next I am going to ask you about the time you spend doing different types of physical activity in a typical week. Please answer these questions even if you do not consider yourself to be a physically active person.
- Think first about the **time you spend doing work.** Think of work as the things that you have to do such as paid or unpaid work, study/training, household chores, harvesting food/crops, fishing or hunting for food, seeking employment. *[Insert other examples if needed].*
- In answering the following questions **'vigorous-intensity activities'** are activities that require hard physical effort and cause large increases in breathing or heart rate, **'moderate-intensity activities'** are activities that require moderate physical effort and cause small increases in breathing or heart rate

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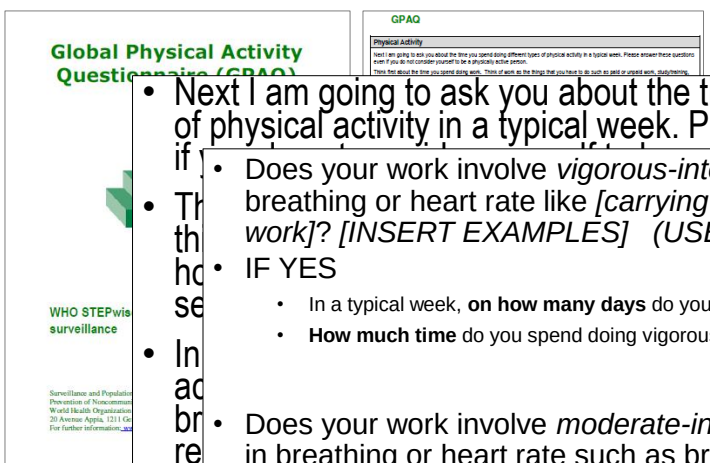
Self report questionnaire: Example GPAQ



- Next I am going to ask you about the time you spend doing different types of physical activity in a typical week. Please answer these questions even if
- Th th ho se
- In ac br re he

- Does your work involve *vigorous-intensity activity* that causes large increases i breathing or heart rate like *[carrying or lifting heavy loads, digging or constructi work]*? *[INSERT EXAMPLES]* *(USE SHOWCARD)* YES/NO
- IF YES
  - In a typical week, **on how many days** do you do vigorous-intensity activities as part of your work?
  - **How much time** do you spend doing vigorous-intensity activities at work on a typical day?
- Does your work involve *moderate-intensity activity*, that causes small increase in breathing or heart rate such as brisk walking *[or carrying light loads]* YES/NO
- IF YES
  - In a typical week, **on how many days** do you do moderate-intensity activities as part of your work?
  - **How much time** do you spend doing moderate-intensity activities at work on a typical day?

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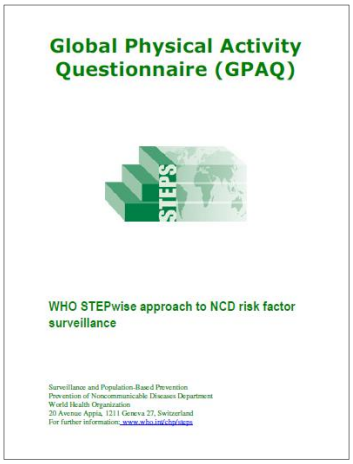
Self report questionnaire: Example GPAQ

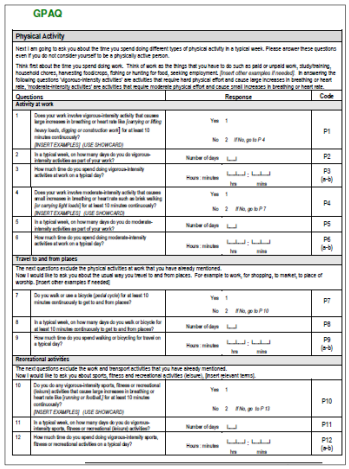


Global Physical Activity Questionnaire (GPAQ)

STEPS

WHO STEPwise approach to NCD risk factor surveillance

Surveillance and Population-Based Prevention
Prevention of Noncommunicable Diseases Department
World Health Organization
20 Avenue Appia, 1211 Geneva 27, Switzerland
For further information: www.who.int/chp/steps



GPAQ

Physical Activity

Next I am going to ask you about the time you spend doing different types of physical activity in a typical week. Please answer these questions even if you do not consider yourself to be a physically active person.

Think first about the time you spend doing work. Think of work as the things that you have to do such as paid or unpaid work, study/training, household chores, harvesting foodcrops, fishing or hunting for food, seeking employment. *[Insert other examples if needed]*. In answering the following questions 'vigorous-intensity activities' are activities that require hard physical effort and cause large increases in breathing or heart rate, 'moderate-intensity activities' are activities that require moderate physical effort and cause small increases in breathing or heart rate.

Questions		Response	Code
Activity at work			
1	Does your work involve vigorous-intensity activity that causes large increases in breathing or heart rate like *[carrying or lifting heavy loads, digging or construction work]* for at least 10 minutes continuously? *[INSERT EXAMPLES] (USE SHOWCARD)*	Yes 1 No 2 *If No, go to P 4*	P1
2	In a typical week, on how many days do you do vigorous-intensity activities as part of your work?	Number of days ☐	P2
3	How much time do you spend doing vigorous-intensity activities at work on a typical day?	Hours : minutes ☐☐ : ☐☐ hrs mins	P3 (h:m)
4	Does your work involve moderate-intensity activity that causes small increases in breathing or heart rate such as brisk walking *[or carrying light loads]* for at least 10 minutes continuously? *[INSERT EXAMPLES] (USE SHOWCARD)*	Yes 1 No 2 *If No, go to P 7*	P4
5	In a typical week, on how many days do you do moderate-intensity activities as part of your work?	Number of days ☐	P5
6	How much time do you spend doing moderate-intensity activities at work on a typical day?	Hours : minutes ☐☐ : ☐☐ hrs mins	P6 (h:m)
Travel to and from places			
The next questions exclude the physical activities at work that you have already mentioned. Now I would like to ask you about the usual way you travel to and from places. For example to work, for shopping, to market, to place of worship. *[Insert other examples if needed]*			
7	Do you walk or use a bicycle (pedal cycle) for at least 10 minutes continuously to get to and from places?	Yes 1 No 2 *If No, go to P 10*	P7
8	In a typical week, on how many days do you walk or bicycle for at least 10 minutes continuously to get to and from places?	Number of days ☐	P8
9	How much time do you spend walking or bicycling for travel on a typical day?	Hours : minutes ☐☐ : ☐☐ hrs mins	P9 (h:m)
Recreational activities			
The next questions exclude the work and transport activities that you have already mentioned. Now I would like to ask you about sports, fitness and recreational activities (leisure). *[Insert relevant terms]*.			
10	Do you do any vigorous-intensity sports, fitness or recreational (leisure) activities that cause large increases in breathing or heart rate like *[running or football]* for at least 10 minutes continuously? *[INSERT EXAMPLES] (USE SHOWCARD)*	Yes 1 No 2 *If No, go to P 13*	P10
11	In a typical week, on how many days do you do vigorous-intensity sports, fitness or recreational (leisure) activities?	Number of days ☐	P11
12	How much time do you spend doing vigorous-intensity sports, fitness or recreational activities on a typical day?	Hours : minutes ☐☐ : ☐☐ hrs mins	P12 (h:m)

- Too many questions - takes too long
- Confusion about terms
- Feels repetitive to responder (and the interviewer)
- Show card examples are useful but also limited
- Difficulty estimating time spent - leads to rounding

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Can we do better?

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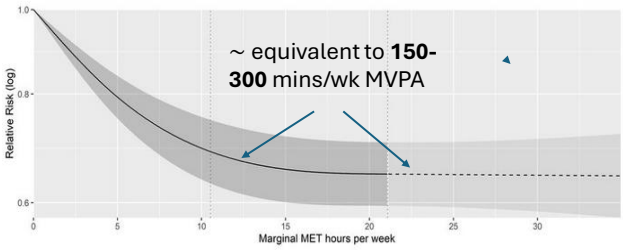
Systematic review

Non-occupational physical activity and risk of cardiovascular disease, cancer and mortality outcomes: a dose–response meta-analysis of large prospective studies

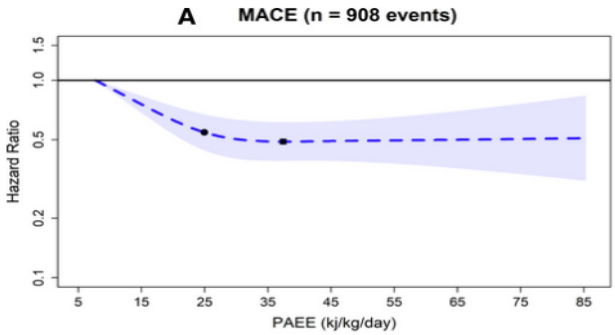
Cardiovascular mortality



(**TOTAL WEEKLY VOLUME OF PHYSICAL ACTIVITY**)

Garcia L, *et al. Br J Sports Med* 2023;**57**:979–989.

Dose-response of physical activity volume (EE) vs. Major Adverse Cardiovascular Events



PHYSICAL ACTIVITY VOLUME (EE)

(Stamatakis E, et al Circulation 2025; 151:1063-75)

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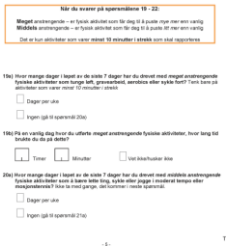
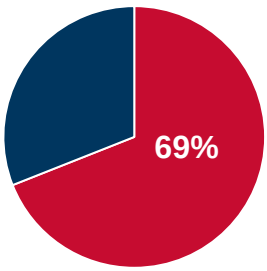
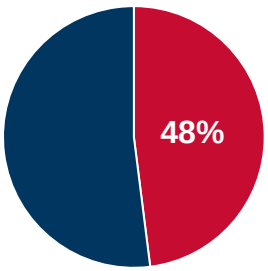
Self report overestimates MVPA in adults







The Norwegian National Physical Activity Survey (NNPAS)

NIH Department of Sports Medicine NORWEGIAN SCHOOL OF SPORT SCIENCES

Thanks to Jostein Steene-Johannessen

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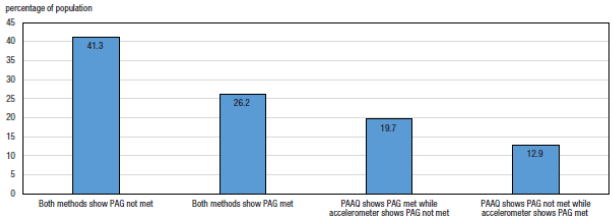
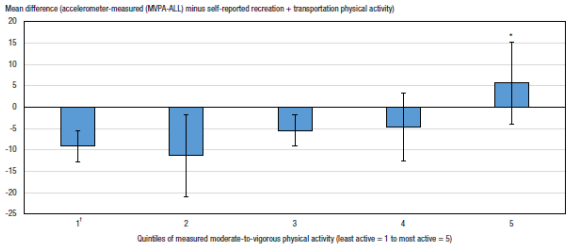
Figure 5
Percentage of population where the physical activity guideline was met by both the questionnaire and the accelerometer, neither the questionnaire nor the accelerometer, the questionnaire only, and the accelerometer only, household population aged 18 to 79, Canada excluding territories, 2014 and 2015

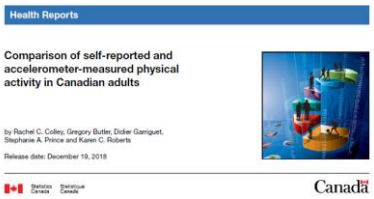


PAG – physical activity guidelines
PAAQ – physical activity adult questionnaire
Source: Canadian Health Measures Survey, Cycle 4 (2014 and 2015).

Figure 3
Mean difference between accelerometer-measured and self-reported physical activity, presented by quintile of physical activity level, household population aged 18 to 79, Canada excluding territories, 2014 and 2015



* significantly different from reference category ($p < 0.05$)
† reference category
MVPA – moderate-to-vigorous physical activity
ALL – all minutes of MVPA included
Source: Canadian Health Measures Survey, Cycle 4 (2014 and 2015).



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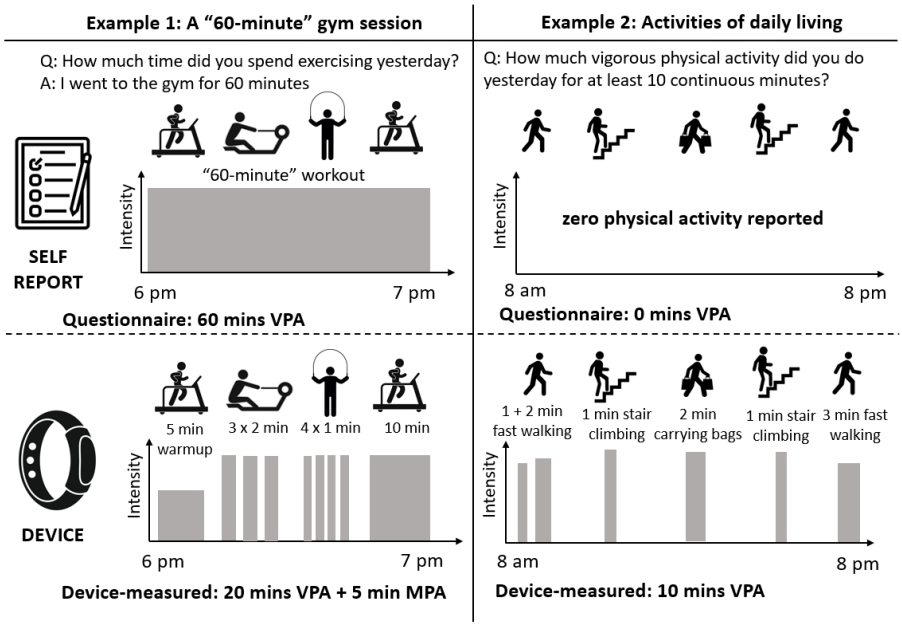
Potential impact of wearables on physical activity guidelines and interventions: opportunities and challenges

Jason MR Gill,[1] Timothy J Chico,[2] Aiden Doherty,[3] Jessilyn Dunn,[4] Ulf Ekelund,[5,6] Peter T Katzmarzyk,[7] Karen Milton,[8] Marie H Murphy,[9] Emmanuel Stamatakis[10]



Gill JMR et al. BJSM 2023 2023;57:1223-1225.

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Deaths potentially averted by small changes in physical activity and sedentary time: an individual participant data meta-analysis of prospective cohort studies

Ulf Ekelund, Jakob Tarp, Ding Ding, Miguel Adriano Sanchez-Lastra, Knut Eirik Dalene, Sigmund A Anderssen, Jostein Steene-Johannessen, Bjorge H Hansen, Bente Morseth, Laila A Hopstock, Edvard Sagelv, Peter Nordström, Anna Nordström, Maria Hagströmer, Ing-Mari Dohrn, Keith M Diaz, Steven Hooker, Virginia J Howard, I-Min Lee, Morten W Fagerland

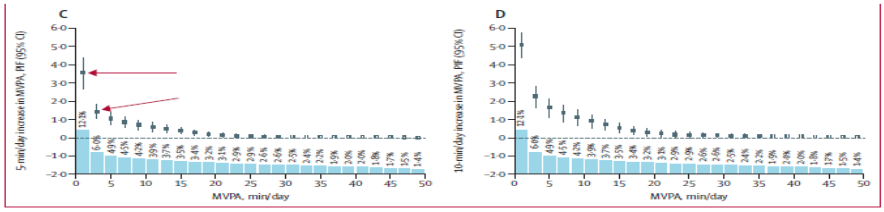

Figure 1: MVPA and mortality risk in the accelerometer consortium
Figure shows results of meta-analysis of seven studies. HRs for mortality for 5-min/day (eg, the HR for a 5-min/day increase from 1 min/day to 6 min/day [red arrow]; A) and 10-min/day (eg, the HR for a 10-min/day increase from 1 min/day to 11 min/day [red arrow]; B) increases in MVPA from observed level. Percentage PIF for 5-min/day (eg, a 5/min/day increase in MVPA in those who accumulated ≤2 min/day of MVPA was associated with 3-6% of preventable deaths [red arrow], and a 5-min increase among those accumulating 2–4 min of MVPA per day was associated with 1-4% of preventable deaths [red arrow]; C) and 10-min/day increase in MVPA from observed level (D). Bar charts show distribution and percentages of participants. All results are based on fully adjusted models (age, sex, accelerometer wear-time, BMI, smoking, education, cardiovascular disease, cancer, and diabetes). HR=hazard ratio. MVPA=moderate-to-vigorous-intensity physical activity. PIF=population impact fraction.

THE LANCET

"Small and realistic increases in [physical activity] of 5 min/day might prevent up to 6% of all deaths."

Lancet 2026; 407:339-49

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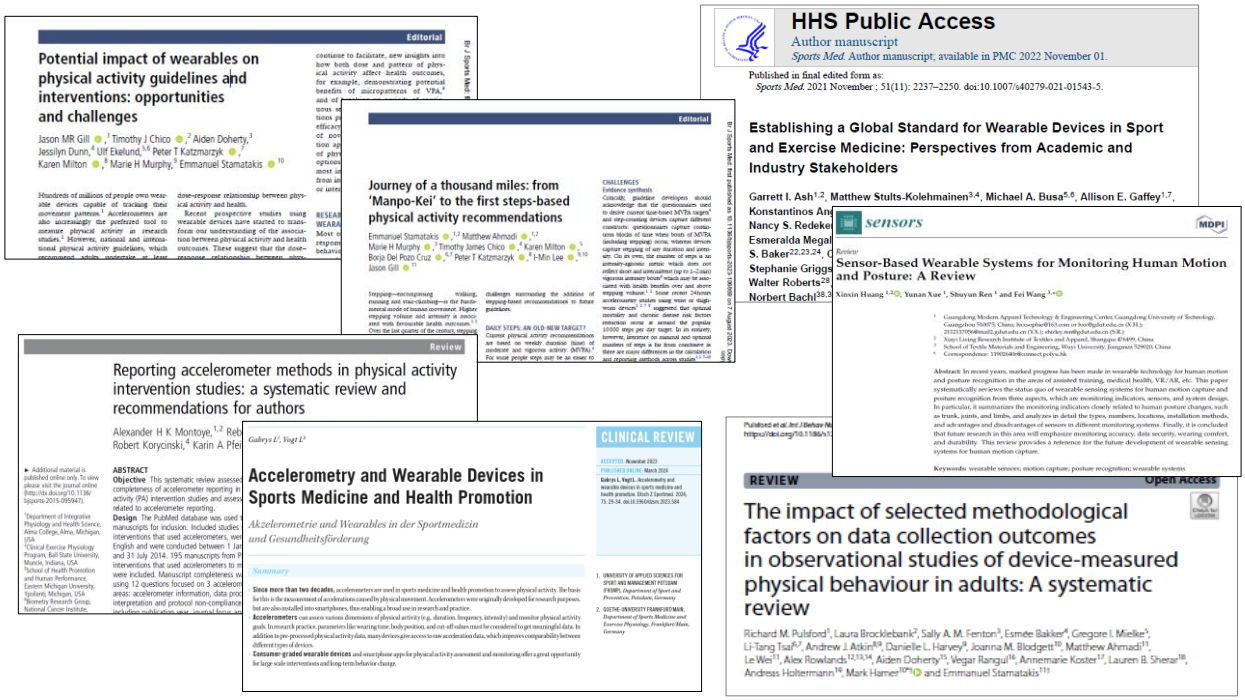

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How many countries are using wearable data in national health monitoring systems?

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Of 194 countries….

About 12

- and some are just at piloting stage / 1st time

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Primary aim of WHO work agenda

To develop global guidance on the use of wearable devices to measure physical activity and sedentary behaviors in national population health monitoring systems in adults and adolescents

In the future….

sleep (when WHO has guidelines on sleep behaviours)

other age group (<10 years; <5 years)

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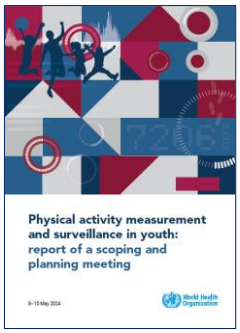
Consultations and expert meetings to advance workplan

1

Nov 2023

2



May 2024

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June 2025

4

Nov 2025

4th meeting report FORTHCOMING

Acknowledgements: To the participants, contributors and funding organizations

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Consultations and expert meetings to advance workplan

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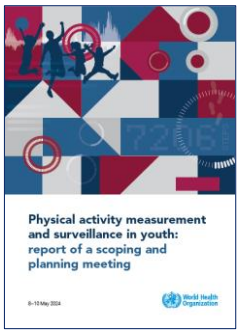
Nov 2023

2



May 2024

- Self-report measurement has persistent and serious limitations ***≠ fit for the future?***
- Rapid advances in sensors, wearables and analytics **= *very promising***
- Extensive use in cohort research studies **=** *proof of use at scale [and need to align]*
- **BUT** - uncertainty and issues

  device choice, methods, feasibility, cost, system integration, comparability

  = need to look into this more….

Acknowledgements: To the participants, contributors and funding organizations

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June 2025

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3

June 2025

Meeting Objectives / Outcomes

1. Agreed a core set of primary output metrics
2. Develop draft device specifications = required features to be "fit for purpose" (& mindful of cost)
3. Review availability and evidence of performance of available open-source algorithms
4. Discuss applicability to older adult populations and diverse populations

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Three core metrics - to focus guidance on initially

1. **Moderate-to-vigorous physical activity (MVPA) (mins)**
2. **Step count – (per day)**
3. **Sedentary behaviour (mins)**

At this time:

- *Sleep and muscle strengthening out of scope*

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Device specifications (and protocols): *DRAFT at a glance table*

Wear protocols summary	
2.1 Wear location	Wrist worn – non-dominant AND Recommended to record the wear location for each responder as some individuals may decline to wear on non-dominant wrist
2.2 Wear time	Wear days: Minimum of 7 days / Optional: collect >7 days Wear time: No decision
2.3 Device orientation	Specification of non-dominant wrist
Device properties including sensor specifications summary	
3.1 Number of axes	Minimum: Triaxial
3.2 Dynamic range	Minimum: +/- 6g
3.3 Resolution	Minimum: 12 bits – 3 milli-g (milli-gravity units)
3.4. Sensitivity	A Least Significant Bit (LSB) of ~0.003
3.5 Noise Floor	Minimum: 1-5 milli-g
3.6 Temperature sensitivity	Minimum: operational range should be minus 20°C to plus 50°C
3.7 Sampling frequency	Minimum: 30Hz Optional – 40Hz or higher
3.8 Memory capacity	Minimum: Memory capacity must be sufficient to support data collection at the specified number of days, hours per day and sample frequency AND Raw accelerometer data must be stored in a non-volatile was (to avoid data loss should battery life end)
3.9 Battery life	Minimum: Battery needs to be sufficient to support the required data collection days and hours per day at specified sample frequency AND Preferably the device should not require the user to re-charge during the data-collection period AND If the device does require re-charging there must be mechanism(s) to protect the data and avoid user access or user download
3.10 Battery recharging and replacement	Minimum: Device must be re-usable and re-chargeable by central coordinator to enable re-use within the surveillance programme AND Devices must be re-chargeable at scale (i.e., with use of a docking station for large batch recharging)
3.11 Feedback to user	Provision of feedback on physical activity/sedentary: No feedback should be provided to user on physical activity (steps, minutes or other metrics) or on sedentary behaviours/time Provision of other feedback or device uses: No decision
3.12 Use of multi-sensor approach	Minimum: (i) Battery health and (ii) Temperature Optional – Temperature and battery health sensors can be switched on/off depending on country context. OTHER SENSORS [to assist analysis/ other purposes/research] (i) Heart rate (via PPG) – no decision (ii) Global positioning systems (GPS) – exclude (not required) (ii) Gyroscope – optional – as this has significant impact on battery requirements
3.12 Waterproof and dustproof and intended for prolonged skin contact	Minimum: (i) Water resistant (IPx8) (ii) Dust tight (IP6x) (iii) materials certified as hypoallergenic and biocompatible and intended for prolonged skin contact
3.13 Band fitting	Minimum: Wrist band must be adjustable, removable and cleanable or disposable (and replacement units available)
3.14 Data handling and transfer	No decision

Processing protocols	
4.1 Access to raw signal	Minimum: Raw accelerometer data must be accessible
4.2 Frequency filtering	Minimum: 20 Hz Butterworth filter
4.3 Epoch length	Maximum: 60 seconds Optional 10 sec or 30 seconds – not decided
4.4 Minimum bout length and interruption tolerance	Use of minimum length of time (i.e., defining a "bout"): Not required/recommended to use in data analysis. Tolerance for interruptions: No decisions
4.5 Identifying non-wear	No decisions
4.6 Minimum wear-time inclusion criteria	No decisions

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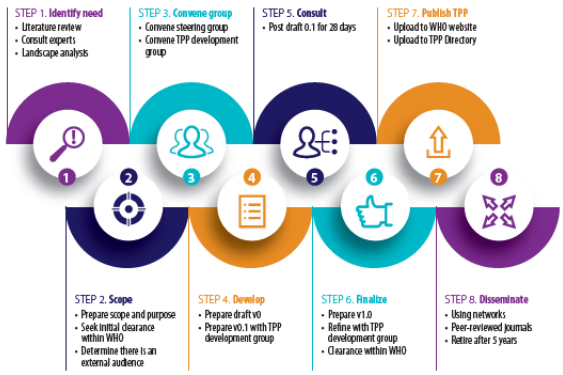
And producing WHO Technical Product Profile (TPP)

Well defined stepped process to develop an 'end-to end' user guide for a product (also service / drug/ treatment)

Fig. 1. Outline of TPP development process



Note: The WHO Research for Health (RFH) department is available for support at any step of the process.

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Understanding the "wearable" device landscape

What is out there ?

- **Wearable Landscape Network**
- an open science initiative committed to fostering collaboration in the field of wearable technologies.
- bringing together a wide range of expertise, to develop innovative solutions - address methodological and content-specific challenges in the field.

- https://www.wearable-landscape.info/vision

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Meeting Objectives

1. Review country experiences within national systems
2. Examine evidence on wrist-worn measurement of cycling
3. Identify priority actions for integration into STEPS and national systems
4. Revisit standardization - what is needed and what is not?

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What did we learn?

Different national models, different aims

- Estimating prevalence
- Understanding patterns of movement
- Gaining insights to inform policy and behaviour change interventions

- Use wearable
  - in parallel to questionnaire
  - as replacement to questionnaire

= Tension is not "device vs self-report" but how to combine them effectively

Representativeness = dominant challenge

- Who is typically missed?
  - Older adults
  - Lower socioeconomic groups
  - Migrants
  - People with poorer health/functional limitations

  = bias becomes a health equity issue
- Risk of bias even with large sample size
- Shared experiences but effective mitigation strategies remains limited

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SUMMARY TABLE 1 COUNTRY SYSTEM/STUDY OVERVIEW

	1.1 SYSTEM / STUDY NAME	3.1 DEVICE	3.4 WEAR LOCATION	1.6 RECENT YR	1.5 FIRST YR	1.7 # TIMES	1.4 Partners	3.3 Device Procurement	4.2 (sub)Sample (initial[1])
Canada	Canadian Health Measures Survey (CHMS)	Actigraph wGT3x-BT	Waist/hip	2025-27	2007	8 cycles	None	purchased	3,264
Denmark 1	Danish National Health Survey (DNHS)	SENSmotion	Thigh	2023	2023	1	Express/AS Sensmotion	Purchased	6,993
Denmark 2	Moving Denmark 2020	Axivity AX3	Thigh	2021/22	2021/22	1	University	Purchased	1,248
Finland 1	Healthy Finland 2022-23	Actigraph GT9X Link	Wrist, ND*	2023	2017	2	University	Purchased	1,980
Finland 2	FINFIT STUDY	UKK- RM42	Waist, ND*+ Wrist at night	2025-26	2017-18	3	None	Own device	10,500
Japan	National Health and Nutrition Survey (NHNS-J)	Yamax (pedometer)	Waist/hip (front pocket from 2023	2024	1989	34	Yamasa Corp NIBN	Purchased	4,272
Norway	Norwegian National PA Surveillance (NNPAS)	Actigraph GTX3X+BT	Waist/hip	2022/22	2008/09	3	Ministry & University & NIPH	Purchased	6,089
Sweden	Health on Equal Terms (HLV)	UKK - RM42	Waist, ND*+ Wrist at night	2025	2018	2	Ministry&UKK + others	Borrowed	4,000
USA	National Health & Nutrition Examination Surv (NHANES)	Actigraph GT3x	Wrist – ND*	2011/14	2003/06	2	NIH, CDC	Purchased	16,000
Netherlands	Accelerometer pilot	ActivPAL 3/4	Thigh	2024	2023	pilot	RIVM, CBS, ++	Purchased	1,100

1 Actical device (waist worn) used for cycles 1- 6 – changed to Actigraph in cycle 7 in 2022 *ND non dominant wrist [1] rough indication of initial/starting device sample

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SUMMARY TABLE 3 REQUIREMENTS AND TRAINING (AND INDICATIVE SAMPLE SIZE)

	1.1 SYSTEM / STUDY NAME	3.1 DEVICE	2.8 Wear Time	6.11 Recharge required	3.5 Other sensors?	5.5. Device problems %	6.2 Data download	8.1 Admin/Data Training?
Canada	Canadian Health Measures Survey (CHMS)	Actigraph wGT3x-BT	7 /24	No	Gyroscope PPG	Malfunction: 2%	StatCan Secure Server	Admin-Y Data- N
Denmark 1	Danish National Health Survey (DNHS)	SENSmotion	7 /24	No	Gyroscope	Very few	Amazon webserver	Admin-Y Data- N
Denmark 2	Moving Denmark 2020	Axivity AX3	7 /24 -RW*	No	Skin sensor	Minor – few	University server	Admin-Y Data- N
Finland 1	Healthy Finland 2022-23	Actigraph GT9X Link	7 /24 -RW*	No	No	Battery (COVID)	Ssecure server in THL	Admin-N Data- N
Finland 2	FINFIT STUDY	UKK- RM42	7 /24 -RW* + switch**	No	Y - posture metrics	None	UKK server	Admin-Y Data- N
Japan	National Health and Nutrition Survey (NHNS-J)	Yamax (pedometer)	1 Day waking hrs	No	No	No info	Manual diary data entry	Admin-Y Data- N
Norway	Norwegian National PA Surveillance (NNPAS)	Actigraph GTX3X+BT	7/waking hr only - RW*	No	No	2-3%	University server	Admin-Y Data- N
Sweden	Health on Equal Terms	UKK - RM42	7 /24 -RW* + switch**	No	Y - posture metrics	Few	University&PH agency	Admin-N Data- N
USA	NHANES	Actigraph GT3x	7 /24	No	Light sensor	Some detected on return	Contractor & NCHS	Admin-Y Data- Y
The Netherlands	Accelerometer pilot	ActivPAL 3/4	9 days / 24 RW*	Yes	Gyroscope PPG	Some battery issues	At survey center & RIVM	Admin-Y Data- N

*RW-Instructed to Remove for Water based/shower/bath; **Switch = instruction for different wear location at night;

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What did we learn?

What to standardize — and what not?

- **Algorithms:** yes – critical
- **Device wear location**: a critical choice – but solutions emerging
- **Device choice:** less important
- **Wear protocols:** yes - set minimum

AND

- **Device function specifications**: yes – set minimum and optimal

LET'S Be active Everywhere Everyone Everyday

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So, what next?

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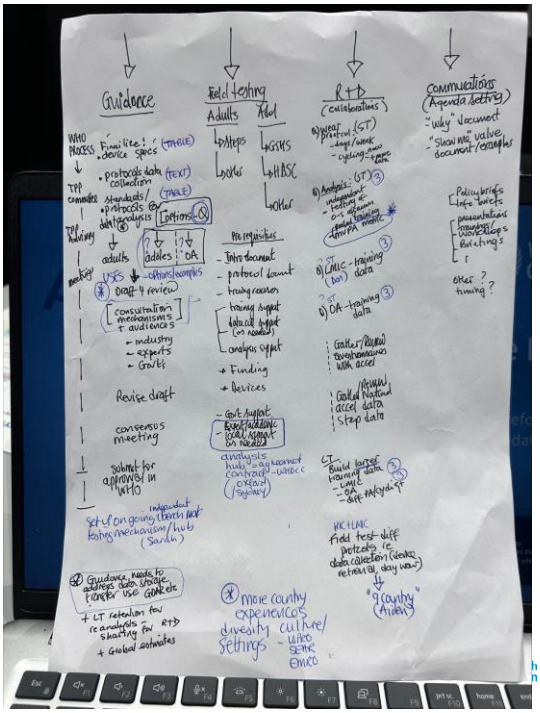
Under development and underway

1. Guidance (v1)
2. Research / knowledge gaps
3. Field implementation “trial/test/adapt”
4. Comms, advocacy & policy work

AND resource mobilisation



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RESEARCH
Global Research Consortium: Self formed and active !

- **WAVES** - Wrist Algorithms: Verification, Evaluation and Surveillance initiative

- Daniel Fuller - University of Saskatchewan
- Aiden Doherty – University of Oxford
- Laura Brocklebank – University of Oxford
- Ben Maylor – University Oxford
- Matthew Ahamdi – University of Sydney
- Manos Stamatakis – University of Sydney
- Kerstin Back – Norwegian University of Science and Technology (NTNU)
- Sarah Kozey Keadle – California Polytechnic State University
- Chuck Matthews – NIH, USA
- Julian Martinez – NIH, USA
- Scott Strath – University of Wisconsin-Milwaukee
- Fiona Bull - WHO

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A framework for evaluating devices to measure physical behavior

Refine and optimize; informed by time use data

Phase 0	Phase I	Phase II	Phase III	Phase IV
Reliability	**Development**	**Preliminary testing**	**Efficacy**	**Adoption**
Testing reliability of device signal Mechanical shaker, comparison of raw signal (MAD, ENMO)	Lab-based calibration Fixed start/end times Known activities	Semi-structured protocol Includes some transitions between activities	Naturalistic, representative activities Independent sample Reference measures	Implement algorithm into applied study (e.g., intervention, cohort, surveillance)

CAL POLY

Keadle et al., A Framework for Evaluating Wearable Devices. ESSR 2019

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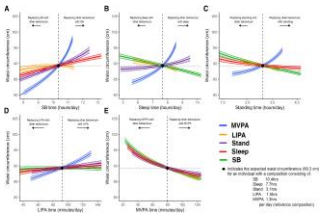
Prospective Physical Activity, Sitting & Sleep consortium (ProPASS)

An international research platform of wearable-based cohort studies aimed at understanding the health effects of physical activity, posture and sleep.

June 2025: >45 cohort studies; n ~>250K from 20 countries include 4 LMICs

Harmonization of 24-hour thigh and wrist-worn wearables

Secured remote federated data access and analysis via DataSHIELD



Mission:
Spearheading the transition to device-based physical activity guidelines

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- **Complete global guidance - based on 'good enough'**
- **Conduct demonstration studies - "trial and adapt"**
- **Address knowledge gaps with targeted research – algorithm, training data, LMIC**
- **Govt engagement and policy-facing communications**
- **Resource mobilisation and investment**

- A continued role for WHO in convening, policy alignment, technical stewardship, and advocacy for advancing wearable agenda in national health surveillance systems

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“Let us not let perfection get in the way of starting - it is time for this transformational leap in surveillance methods”

Meeting Delegate

Thank you

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